

# What Makes People Happy? A Bayesian Hierarchical Analysis of Life Satisfaction Across 66 Countries

## Abstract

This paper examines the individual and cross-national determinants of subjective life satisfaction using Wave 7 of the World Values Survey, a nationally representative sample of over 90,000 respondents across 66 countries. We fit a sequence of Bayesian hierarchical linear models via Hamiltonian Monte Carlo sampling, progressing from complete pooling to random intercepts and random slopes, with model comparison conducted using leave-one-out cross-validation. Results indicate that individual-level health (posterior mean 0.584) and income (posterior mean 0.394) are the two strongest predictors of life satisfaction, while the effect of education diminishes once income is controlled for. Beyond individual characteristics, national context accounts for an estimated 11.5% of the total variation in reported wellbeing, variation not captured by a non-hierarchical model. The income-satisfaction relationship itself is not uniform across countries: the estimated income effect ranges from near zero in some countries to over three times the global average in others, reflecting broader differences in material circumstances and social context. These findings underscore the importance of accounting for multilevel structure when modeling subjective wellbeing across what are diverse national settings.

## 1 Introduction

The question of what determines happiness is one that scientists have sought to answer for a very long time. Despite this, this question still remains empirically open. Someone's subjective wellbeing, in other words, how satisfied people feel about their lives, varies greatly across both individuals and countries. Understanding the sources of this variation has extreme relevance in both scientific and policy-related fields. For example, if income is the dominant driver of life satisfaction, it will likely have much different implications than if health or national culture were the primary predictor.

Cross-national survey data offers a powerful tool for addressing this question, as it allows us to examine individual-level predictors of wellbeing while simultaneously accounting for the country context in which people live. The nested structure of data containing information about individuals within countries calls for a hierarchical modeling approach, which allows effects to vary across groups while pooling information across them to produce stable estimates.

In this paper, we use Wave 7 of the World Values Survey (WVS), collected between 2017 and 2022 across 66 countries and over 90,000 respondents, to examine the determinants of life satisfaction. We fit a sequence of Bayesian hierarchical linear models of increasing complexity, comparing random intercepts, and random intercepts with random slopes for income. Model comparison by way of leave-one-out cross-validation (LOO-CV) is used to formally evaluate whether the added complexity of each model is warranted by the data in question.

We address two scientific questions. First, which individual-level characteristics predict life satisfaction, and how strong are those effects? Second, after accounting for individual characteristics, how much of the remaining variation in life satisfaction is attributable to national context, and does the effect of income on satisfaction vary across countries?

## 2 Analysis and Results

### 2.1 Data

The data for this analysis comes from Wave 7 of the World Values Survey (WVS), a cross-national social survey conducted between 2017 and 2022 (1). The WVS is one of the longest-running and most comprehensive surveys on human values and beliefs. In Wave 7, the survey covers nationally representative samples across 66 countries. For each country, approximately 1,000 to 2,000 participants were interviewed on topics ranging from their economic stability to personal wellbeing.

Our outcome variable is self-reported life satisfaction, measured on a 1 to 10 scale in response to the question “All things considered, how satisfied are you with your life as a whole these days?” We model this as a continuous outcome, a standard practice in the wellbeing literature (2, 3), while acknowledging that the scale is technically ordinal. The data includes several predictor variables; a subset of these was chosen to be used in this analysis. Individual-level predictors include household income decile (Q288, 1–10, country-specific scale), highest educational attainment (Q275, ISCED 2011, treated as ordinal), self-rated

health (Q47P, 1–5, where 1 = very poor and 5 = very good), age in years (Q262), social trust (Q57P, binary: “most people can be trusted” vs. “need to be very careful”), and gender (Q260, binary).

After removing participants with missing values on any of these variables, 90,940 complete cases remained across 66 countries. To make computation manageable for the Bayesian hierarchical models, we drew a stratified random sample of 150 participants from each country, yielding a working dataset of 9,900 observations. With 150 respondents per country, each country-level estimate is based on a sample size large enough to estimate a mean with reasonable precision (a standard error of roughly  $2.0/\sqrt{150} \approx 0.16$  on the 10-point satisfaction scale, given the observed residual standard deviation of approximately 2.0). This sample size was chosen as a balance between computational tractability and statistical precision. All continuous predictors were standardized to have mean zero and unit variance prior to modeling so that coefficient estimates are directly comparable in magnitude.

**Figure 1** shows the distribution of life satisfaction across all respondents. The distribution is left-skewed with a mean of 7.1 and a mode of 8, consistent with the well-documented positivity bias in self-reported wellbeing (3). The pronounced spike at 10 reflects a ceiling effect common to bounded satisfaction scales.



Figure 1: Distribution of life satisfaction across all 9,900 respondents.

## 2.2 Exploratory Data Analysis

Before fitting the models, we examine the structure of the data to understand the distribution of the outcome variable, the relationships between predictors and life satisfaction, and the degree of variation across countries. This last point is specifically important, as it motivates the hierarchical modeling approach taken in Section 2.3.

**Figure 2** shows the relationship between income decile and mean life satisfaction, aggregated across all countries. There is a strong, linear positive relationship, as respondents in the highest income decile report satisfaction scores approximately 1.9 points higher on average than those in the lowest decile. This noticeable pattern also motivates the inclusion of income as a key predictor in all three models.

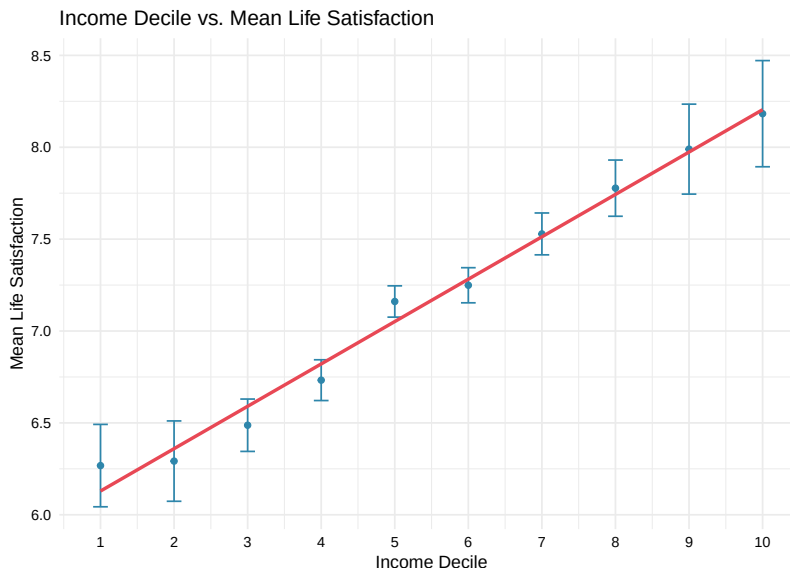


Figure 2: Mean life satisfaction by income decile.

**Figure 3** examines how the income-satisfaction relationship varies across countries. The country-specific regression lines reveal substantial heterogeneity, both in baseline satisfaction levels (intercepts) and in the strength of the income effect (slopes). Certain countries show steep positive income gradients while others show almost flat or slightly negative relationships. The variation in slopes is the empirical motivation for Model 3, which allows the income effect to vary across countries.

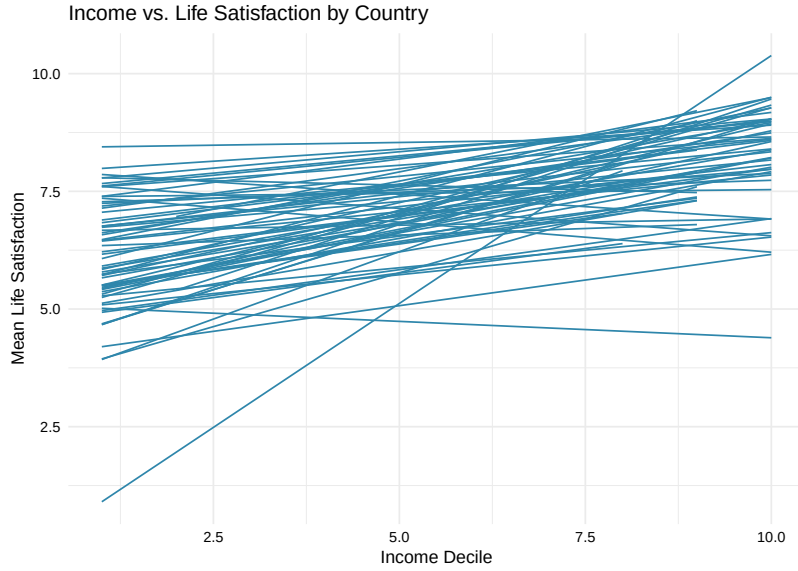


Figure 3: Country-specific income-satisfaction regression lines.

**Figure 4** shows the respective relationships between social trust and life satisfaction, and health and life satisfaction. Social trust shows a very modest positive effect, as respondents who believe most people can be trusted reported around 0.4 points higher life satisfaction on average. Self-rated health shows a much stronger association, as respondents who reported very good health (5) scored nearly 3 points higher in life satisfaction than those who reported very poor health (1).

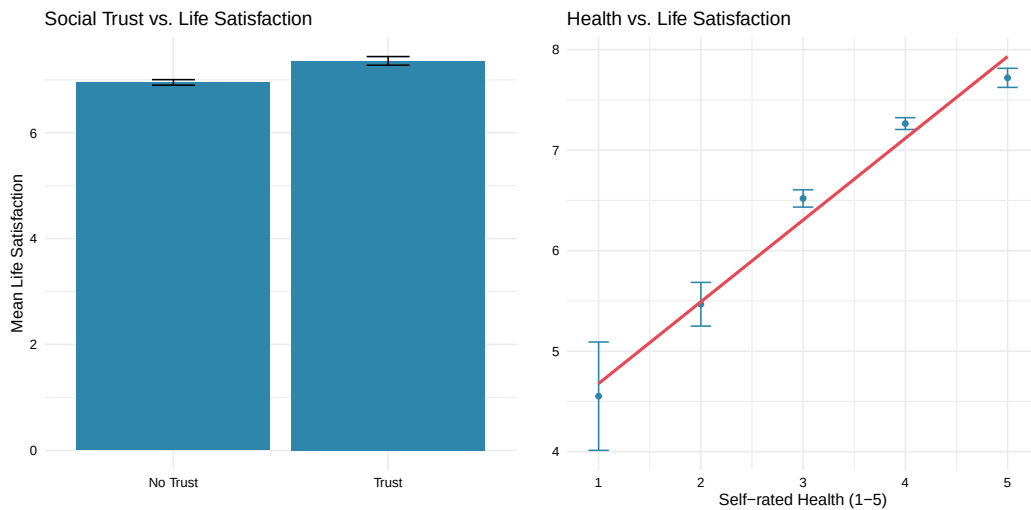


Figure 4: Social trust and self-rated health vs. mean life satisfaction.

Lastly, **Figure 5** displays mean life satisfaction by country in ascending order. The range

spans about 4 points, from Iraq (lowest, approximately 4.2) to Puerto Rico (highest, approximately 8.4). It is clear that national context plays a large role in subjective wellbeing, beyond what individual characteristics alone can explain. This inter-country variation seen here is a primary motivation for the hierarchical structuring of the models used in this analysis.

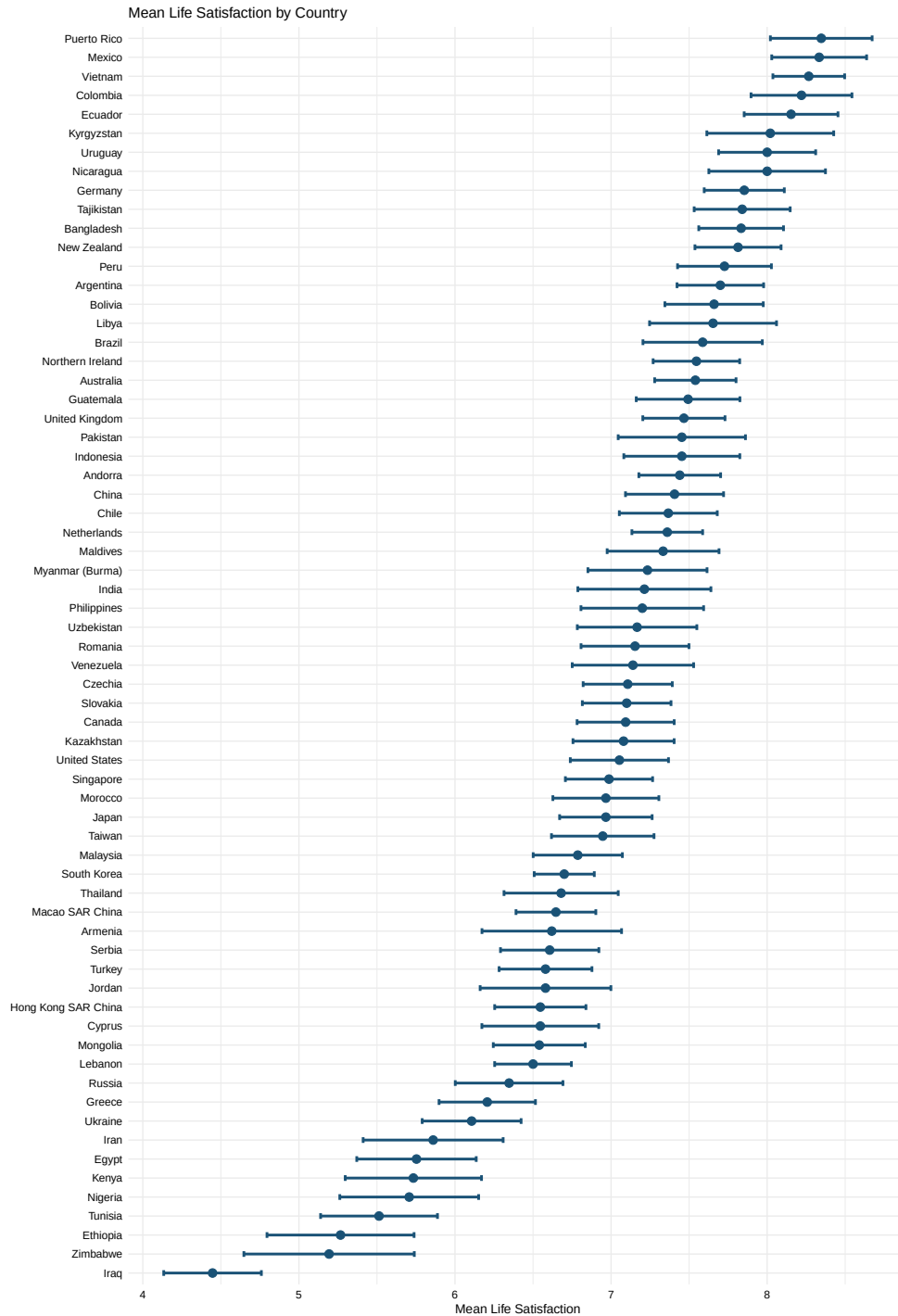


Figure 5: Mean life satisfaction by country.

## 2.3 Model Specification

Three Bayesian models of increasing complexity were fit to examine both individual and cross-national predictors of life satisfaction. Each of these models builds on the last. All three models treat life satisfaction as a continuous outcome with a Gaussian likelihood. The fitting of these models was performed using the **rstanarm** package in R (5), which implements full Hamiltonian Monte Carlo sampling using Stan. Each model was run with 4 chains of 4,000 iterations and 2,000 warmup draws.

**Model 1: Complete Pooling.** The simplest model treats all 9,900 observations as independent, ignoring country structure entirely. This is a standard Bayesian linear regression of the form:

$$y_i = \alpha + \beta_1 \cdot \text{income}_i + \beta_2 \cdot \text{health}_i + \beta_3 \cdot \text{education}_i + \beta_4 \cdot \text{age}_i + \beta_5 \cdot \text{trust}_i + \beta_6 \cdot \text{gender}_i + \epsilon_i$$

where  $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ . This model serves as a baseline but was expected to be inadequate given the large amount of inter-country variation observed in Section 2.2.

**Model 2: Random Intercepts.** Model 2 introduces a hierarchical structure by allowing every country its own baseline level of satisfaction:

$$y_{ij} = \alpha_j + \beta_1 \cdot \text{income}_{ij} + \beta_2 \cdot \text{health}_{ij} + \beta_3 \cdot \text{education}_{ij} + \beta_4 \cdot \text{age}_{ij} + \beta_5 \cdot \text{trust}_{ij} + \beta_6 \cdot \text{gender}_{ij} + \epsilon_{ij}$$

where  $y_{ij}$  is the life satisfaction of respondent  $i$  in country  $j$ , and the country-specific intercepts are modeled as draws from a common distribution:  $\alpha_j \sim \mathcal{N}(\mu_\alpha, \tau^2)$  (4).

**Model 3: Random Intercepts and Random Slopes.** Model 3 allows the effect of income to vary across countries, motivated by the heterogeneity in slopes observed in **Figure 3**:

$$y_{ij} = \alpha_j + \beta_{1j} \cdot \text{income}_{ij} + \beta_2 \cdot \text{health}_{ij} + \beta_3 \cdot \text{education}_{ij} + \beta_4 \cdot \text{age}_{ij} + \beta_5 \cdot \text{trust}_{ij} + \beta_6 \cdot \text{gender}_{ij} + \epsilon_{ij}$$

The country-specific intercepts and income slopes are modeled jointly as draws from a bivariate normal distribution:

$$\begin{pmatrix} \alpha_j \\ \beta_{1j} \end{pmatrix} \sim \mathcal{N} \left( \begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \Sigma \right)$$

where  $\Sigma$  captures both the variance of country-level intercepts and slopes and the correlation between them. The remaining predictors enter as fixed effects, as there was no strong prior reason to expect their effects to vary meaningfully enough across countries.

**Priors.** Weakly informative priors were placed on all parameters. A normal(7, 2) prior was placed on the global intercept, consistent with the known range of life satisfaction scores, and normal(0, 2.5) priors were placed on all fixed effect coefficients. The residual standard deviation  $\sigma$  received an exponential(1) prior, adjusted by `rstanarm` to exponential(0.44) based on the scale of the outcome. For the covariance structure of the random effects in Model 3, the `decov` prior was used with regularization parameter  $\eta = 1$ , concentration parameter  $\alpha = 1$ , and shape and scale both equal to 1. The `decov` prior decomposes the covariance matrix  $\Sigma$  into a correlation matrix and a vector of variances. An LKJ prior is placed on the correlation matrix, which shrinks the estimated correlation between random intercepts and slopes toward zero and prevents overfitting of the covariance structure (4, 5).

## 2.4 Model Comparison

The three models were compared using leave-one-out cross-validation (LOO-CV), implemented via the `loo` package in R (6). LOO-CV estimates each model’s out-of-sample predictive accuracy by approximating the expected log pointwise predictive density (ELPD). A higher ELPD indicates better predictive performance, and vice versa. The differences in ELPD between each model, along with their standard errors, are reported in Table 1.

| Model                               | ELPD Difference | SE   |
|-------------------------------------|-----------------|------|
| Model 3: Random intercepts + slopes | 0.0             | 0.0  |
| Model 2: Random intercepts          | −39.8           | 11.2 |
| Model 1: Complete pooling           | −621.6          | 38.8 |

Table 1: LOO-CV model comparison. ELPD differences are relative to the best-fitting model (Model 3).

Model 1 performs far worse than the hierarchical models, with an ELPD difference of −621.6 (SE = 38.8). This confirms that ignoring country-level variation produces a fundamentally inadequate model. Model 2 improves substantially over Model 1, but Model 3 outperforms



it by 39.8 ELPD points ( $SE = 11.2$ ). This tells us that allowing the income effect to vary across countries captures meaningful structure in the data that the previous two models miss. As a result, Model 3 is chosen as the final model for all following inference.

## 2.5 Fixed Effects

Table 2 reports the posterior means and 95% credibility intervals for all fixed effects from Model 3. **Figure 6** displays these estimates visually. All predictors are standardized, so the coefficients reflect the change in life satisfaction associated with a one standard deviation increase in each predictor.

| Predictor       | Posterior Mean | 95% CrI Lower | 95% CrI Upper |
|-----------------|----------------|---------------|---------------|
| Income          | 0.394          | 0.315         | 0.472         |
| Education       | -0.029         | -0.076        | 0.019         |
| Health          | 0.584          | 0.542         | 0.627         |
| Age             | 0.155          | 0.110         | 0.200         |
| Social Trust    | 0.139          | 0.036         | 0.240         |
| Gender (Female) | 0.153          | 0.073         | 0.234         |

Table 2: Posterior estimates of fixed effects from Model 3 with 95% credibility intervals.

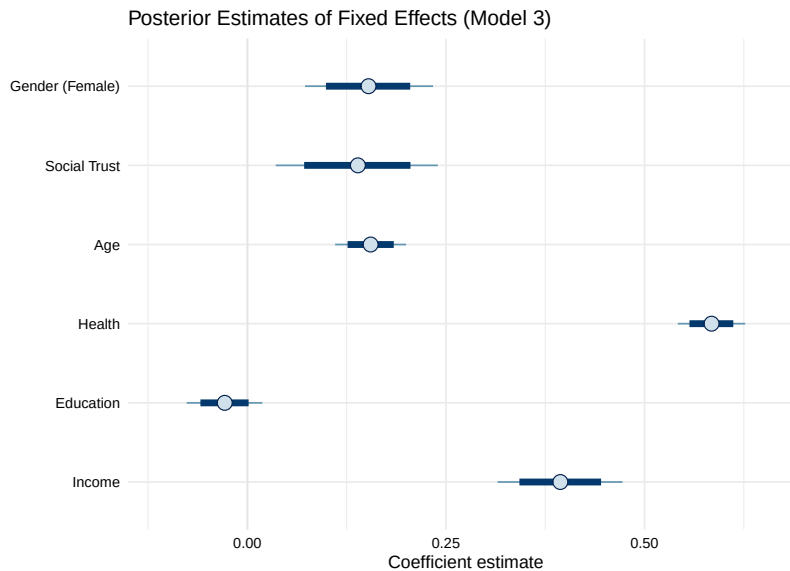


Figure 6: Posterior estimates of fixed effects from Model 3.

Self-rated health emerges as the strongest predictor of life satisfaction, with a posterior mean

of 0.584 per standard deviation increase and a credibility interval entirely above zero. Income is the second strongest predictor at 0.394, also with a credibility interval well beyond zero. We find that these results are largely consistent with cross-national research, which, like our analysis, identifies health and income as the two most dominant predictors of one’s subjective wellbeing (3, 7). It is important to note that health, which is self reported, often expands to include conditions beyond one’s physical condition. The strength of the health effect points to evidence that self-rated health, specifically, captures broader life circumstances beyond health. Issues such as mobility, independence, and social participation, weigh on how satisfied people report feeling about their lives.

Age, social trust, and gender are all credibly positive because their 95% credibility intervals do not include zero, though their overall effects are weaker than self-rated health and income (seen in **Figure 6**). We see that women, as well as respondents who tend to be older and hold more trust in others, report slightly higher life satisfaction on average. The trust effect is consistent with literature connecting social capital to wellbeing, as a sense of trust in others typically reflects greater feelings of social unity (3). The gender effect suggests that women report slightly higher life satisfaction than men on average after controlling for other predictors. This is another pattern that is also studied and observed in other cross-national analyses (7).

Education is the notable exception. Its posterior mean is slightly negative ( $-0.029$ ) and its 95% credibility interval spans zero, meaning its effect cannot be credibly distinguished from having no effect. This result is counterintuitive at first glance, and is well documented in the wellbeing literature. Kristoffersen (2018) argues that education raises reference points and life expectations, so that higher education is associated with greater satisfaction only if the elevated expectations of the education are met (8). Interestingly, when income is already controlled for, as it is here, education’s contribution alone to life satisfaction is relatively insignificant in the grand scheme of things, and may even turn slightly negative as unmet expectations weigh on reported wellbeing.

## 2.6 Country-Level Variation

Beyond the individual-level predictors discussed in Section 2.5, a central question of this analysis is how much of the variation in life satisfaction is attributable to the country a person lives in, and whether certain countries stand out after accounting for their citizens’ individual characteristics. **Figure 7** displays the posterior estimates of the country-level random intercepts from Model 3 (each country’s deviation from the global mean satisfaction of 6.9 after controlling for income, health, education, age, trust, and gender).

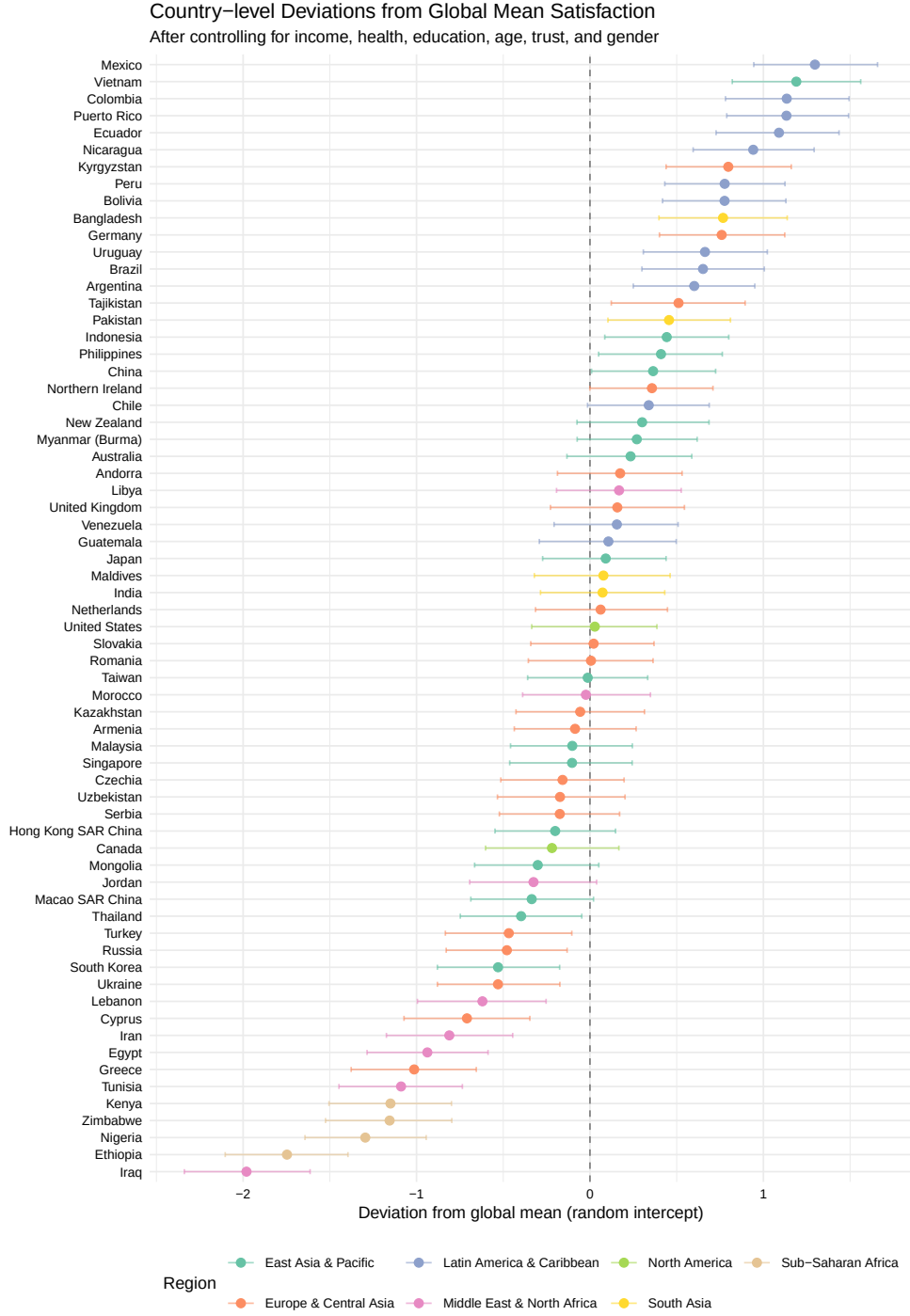


Figure 7: Country-level deviations from global mean satisfaction after controlling for individual predictors.

The estimated variance of the country-level intercepts is  $\hat{\sigma}_\alpha^2 = 0.6$ , indicating that a substantial portion of the total variation in life satisfaction operates at the country level, after controlling for individual characteristics. The intraclass correlation coefficient (ICC

$= \tau^2/(\tau^2 + \sigma^2) = 0.115$ ) indicates that approximately 11.5% of the total variance in life satisfaction is attributable to differences between countries, with the remaining 88.5% reflecting variation among individuals within countries. While modest in absolute terms, this between-country component is substantively meaningful as it represents systematic differences in wellbeing that cannot be explained by any individual-level characteristic. The spread of random intercepts ranges over 3.5 points on a 10-point scale, from around  $-2.2$  (Iraq) to  $+1.4$  (Mexico). This reaffirms the influence that national context has on subjective wellbeing.

At the lower end, we see that Iraq, Ethiopia, Nigeria, and Zimbabwe all show large negative deviations. Iraq’s intercept, as mentioned before, is the most extreme in the entire sample. This is consistent with research documenting the devastating impact of prolonged conflict and instability on population wellbeing in Iraq. Decades of war, economic sanctions, and political violence have created conditions of chronic insecurity that depress life satisfaction well beyond what individual income or health levels would predict (9). As a whole, many countries at this lower end tend to struggle with political instability, conflict, and economic and humanitarian crises.

At the upper end, Latin American countries dominate. Mexico, Colombia, Puerto Rico, Nicaragua, and Ecuador all show strongly positive intercepts. This is extremely interesting as the majority of these countries do not rank highly in income and overall health outcomes. Despite this, they persist at the top. This is a well-documented phenomenon in the wellbeing literature, and is referred to by most as the Latin American happiness paradox. It is that Latin Americans consistently report life satisfaction levels above what their circumstances would predict (10). Though explanations of this phenomenon are not straightforward, researchers typically attribute it to strong family ties and a cultural orientation toward positive affect. This social connectedness compensates for lower material living standards (10).

**Figure 8** shows the country-specific income slopes (the total income effect for each country, combining the global average slope of 0.4 with each country’s deviation). The variation here is striking. Iraq has by far the steepest income slope (approximately 1.5), suggesting that income is an unusually powerful determinant of satisfaction within the country. This makes more sense intuitively in a context of severe material deprivation, where differences in income distribution likely represent the difference between meeting and not meeting basic needs. In contrast, wealthier and more equal countries such as Australia, Japan, and Germany show income slopes close to or below the global average, which is consistent with the diminishing returns to income at higher living standards documented in the broader literature (7).

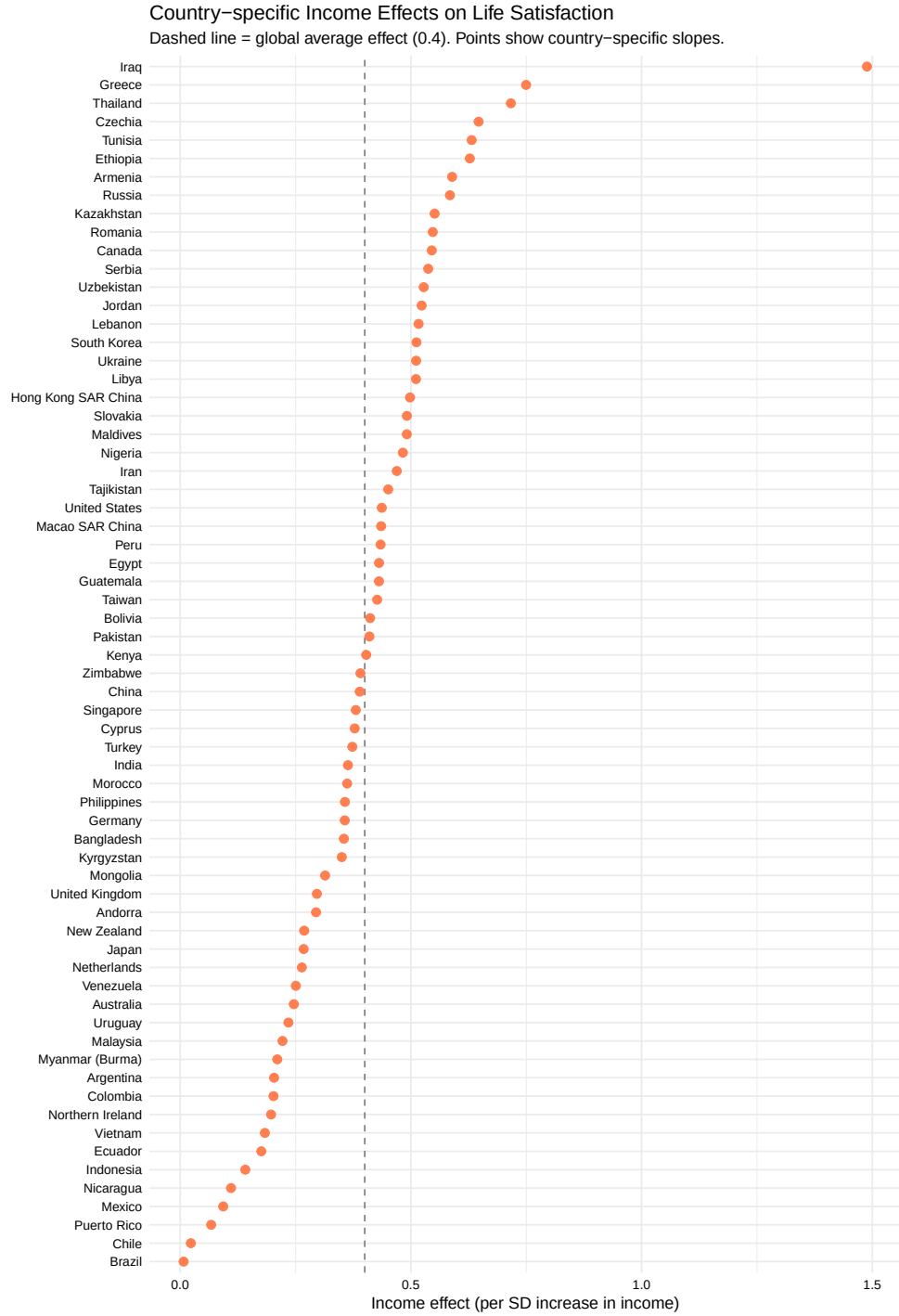


Figure 8: Country-specific income effects on life satisfaction.

The estimated covariance,  $\Sigma[\alpha_j, \beta_{1j}] = -0.107$  (95% CrI:  $-0.177, -0.052$ ), indicates a credibly negative correlation: countries with lower baseline satisfaction tend to show stronger income effects. This pattern, that income matters most where people have the least, is con-

sistent with the broader cross-national literature on the income-wellbeing relationship (7, 10).

## 2.7 Model Diagnostics

**Figure 9** shows the posterior predictive check for Model 3. The light blue lines represent 50 datasets simulated from the fitted model, while the black line shows the observed data. We see that the model captures the overall location, spread, and rough shape of the distribution well, as the replicated datasets are centered at the right mean with appropriate variance. Despite this, there is a notable discrepancy visible, as the observed data has sharp spikes at integer values from 1 through 10. The model, on the other hand, produces smooth continuous predictions. This is an expected consequence of treating a discrete ordinal scale as a continuous Gaussian outcome, and is a well-known limitation of this approach (2). It does not invalidate the model’s inferences about predictor effects and country-level variation, but it does suggest that a fully ordinal model could provide a better description of the outcome distribution in future work.

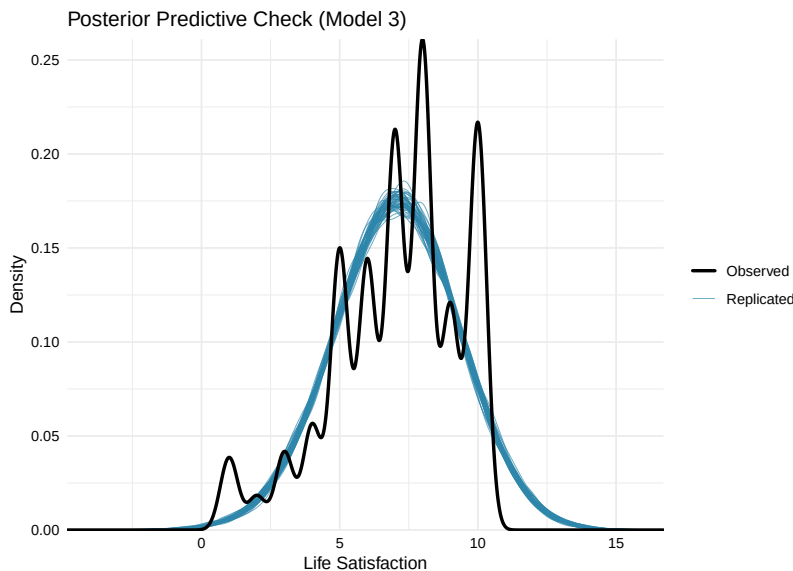


Figure 9: Posterior predictive check for Model 3.

Before drawing conclusions from Model 3, we must assess the reliability of the posterior estimates through MCMC convergence diagnostics. Table 3 reports the  $\hat{R}$  statistic and effective sample size ( $n_{\text{eff}}$ ) for all key parameters. All  $\hat{R}$  values equal 1.0, indicating that the four chains converged to the same posterior distribution. The effective sample sizes range from 634 (intercept) to 12,426 (gender), which all lay well above the minimum threshold of

400 for reliable posterior inference. **Figure 10** displays trace plots for the key parameters, confirming the expected rapid, random fluctuation around a stable mean with no trends. Together, these diagnostics confirm that the posterior estimates reported in Sections 2.5 and 2.6 are reliable.

| Parameter       | $\hat{R}$ | $n_{\text{eff}}$ |
|-----------------|-----------|------------------|
| Intercept       | 1.0       | 634              |
| Income          | 1.0       | 1,397            |
| Education       | 1.0       | 8,448            |
| Health          | 1.0       | 10,641           |
| Age             | 1.0       | 8,670            |
| Social Trust    | 1.0       | 9,678            |
| Gender (Female) | 1.0       | 12,426           |
| Sigma           | 1.0       | 12,302           |

Table 3: MCMC convergence diagnostics for Model 3.  $\hat{R} = 1.0$  indicates full convergence;  $n_{\text{eff}}$  is the effective sample size.

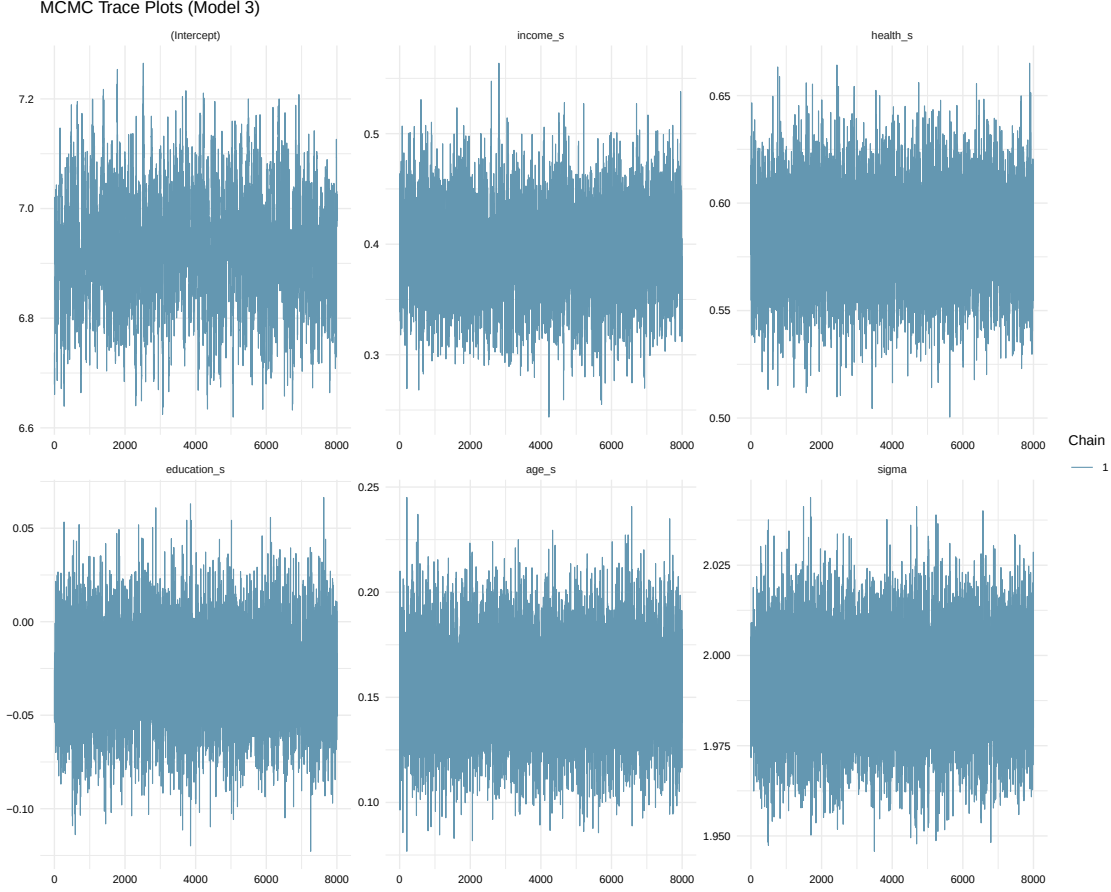


Figure 10: MCMC trace plots for key parameters of Model 3.

### 3 Discussion

The results of this analysis confirm that both individual circumstances and national context independently shape life satisfaction. Health and income are the strongest individual predictors, a finding consistent with the broader literature (3, 7), while education’s near-zero effect once income is controlled for aligns with evidence that its wellbeing returns operate indirectly through material outcomes rather than directly (8).

The country-level results are where the most substantively interesting patterns emerge. The variance of country intercepts ( $\hat{\sigma}_\alpha^2 = 0.6$ ) and the LOO comparison both confirm that national context matters substantially, independent of individual characteristics. The dominance of Latin American countries at the upper end, despite relatively modest material conditions, points to the role of social and cultural factors that our model does not capture, consistent with the Latin American happiness paradox (10). At the lower end, countries marked by conflict and humanitarian crisis cluster together, suggesting that macro-level instability sup-



presses wellbeing in ways that income and health alone cannot offset (9). The variation in income slopes further reinforces that income’s importance is context-dependent, strongest where basic needs are most precarious.

Notably, the negative correlation between country intercepts and income slopes ( $\Sigma[\alpha_j, \beta_{1j}] = -0.107$ , 95% CrI:  $-0.177, -0.052$ ) formalizes the pattern that, generally, income matters most where people are least satisfied. It also suggests that economic development in lower-wellbeing countries could yield greater returns to population happiness compared to equivalent growth in already-satisfied ones. On the other end, wealthier Western countries show that relative income has little bearing on reported satisfaction. Taken together, these findings suggest that the relationship between money and happiness is not fixed, rather, it depends fundamentally on the material context in which people live.

**Limitations.** A few limitations of this analysis are worth acknowledging. First, the outcome variable, life satisfaction on a 1–10 scale, is technically ordinal, and our Gaussian likelihood cannot fully reproduce the discrete and bounded nature of the data, as seen in the posterior predictive check (**Figure 9**). A Bayesian ordinal model via **brms** could address this limitation in future work. Second, although education’s overall effect was not credibly different from zero, we did not test whether this effect, or the baseline differences between countries, could be explained by country-level covariates such as GDP per capita or income inequality. Incorporating predictors like these in future work could help explain why some countries have higher baseline satisfaction than others, and whether education matters more in certain national contexts. Finally, by sampling 150 respondents per country, we sacrifice some statistical power relative to the full dataset, though the consistency of results across models suggests this did not meaningfully affect the conclusions.

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